JAN HUERTAS MARTIN

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RESEARCH PROFILE

My research investigates the physical mechanisms that control chromatin accessibility. Using multiscale molecular simulations integrated with experimental data, I study how nucleosome dynamics, chromatin organization, and protein-protein interactions shape transcription factor search and activity in chromatin. A particular focus of my work is understanding how pioneer transcription factors access nucleosome-occupied DNA and how these mechanisms can be harnessed to improve cellular reprogramming.

ACADEMIC APPOINTMENTS

Independent Research Fellow – Herchel Smith Fellowship **2023-present**
University of Cambridge

Postdoctoral Fellow **2021-2024**
University of Cambridge
HORIZON TMA Marie Skłodowska-Curie Action Postdoctoral Fellow **2022-2024**

EDUCATION

Dr.rer.Nat. in Chemistry (Theoretical Chemistry) **2016-2021**
Max Planck Institute for Molecular Biomedicine - Westfälische Wilhelms Universität Münster (WWU).

M.Sc. in Bioinformatics for Health Sciences **2014-2016**
Universitat Pompeu Fabra, Barcelona

B.Sc. in Human Biology **2010-2014**
Universitat Pompeu Fabra, Barcelona

SELECTED PUBLICATIONS

★ *First/Co-First author*. © *Computational Modelling Lead*. § *Co-corresponding author*

1. ★ **Huertas J**, Maristany MJ, Collepardo-Guevara R. **Oct4 clusters promote DNA accessibility by enhancing chromatin plasticity.** ***BioRxiv***. ***Under Review in Nature Communications*** (2026)
2. © Zhou H, **Huertas J**, [...] Collepardo-Guevara R, Rosen MK. **Multi-scale structure of chromatin condensates rationalizes phase separation and material properties.** ***Science*** (2025)
3. ★ **Huertas J and** MacCarthy CM, […] Schöler HR, Cojocaru V. **OCT4 interprets and enhances nucleosome flexibility.** ***Nucleic Acids Research*** (2022)
4. § Maristany MJ and Perez-Lopez JI, Farr SE, **Huertas J** and Collepardo-Guevara R. **A Goldilocks zone of DNA flexibility defines stable yet plastic nucleosomes, tuned by histone chemistry.** ***BioRxiv. Under Review in Nature Communications*** (2026)
5. © Shimazoe MA, **Huertas J,** [...] Collepardo-Guevara R and Maeshima K. **Linker histone H1 functions as a liquid-like glue to organize chromatin in living human cells.** ***Science Advances*** (2026)

FULL LIST OF PUBLICATIONS

★ *First/Co-First author*. © *Computational Modelling Lead*. § *Co-corresponding author*

6. Maristany MJ, **Huertas J**, et al. **Multiscale simulations of chromatin structure and chromatin phase transitions.** ***Springer Nature, Methods in Molecular Biology*** – in Press (2026).
7. ★ **Huertas J,** Maristany MJ, Perez Lopez JI, Collepardo R. **Pushing the experimental resolution boundary in chromatin fibre organisation using multiscale simulations.** ***Current Opinion in Genetics and Development*** – in Press (2026)
8. © Spicer MFD, Wijma S, Schütte N, **Huertas J** et al. **Electrostatic control of chromatin compaction safeguards against apoptotic DNA release.** ***BioRxiv*** (2026)
9. Li H and Dalgliesh JLT, Lister G, Maristany MJ, **Huertas J** et al. **Mapping chromatin structure at base-pair resolution unveils a unified model of cis-regulatory element interactions.** ***Cell*** (2025)

10. Russel K, […] **Huertas J,** Orozco M, Rosen MK, Collepardo-Guevara R. **Near-atomistic simulations reveal the molecular principles that control chromatin structure and phase separation.** ***BioRxiv*** (2025)
11. Bell C and Chen L and Maristany MJ**,** Blaukopt C, Zhou H, **Huertas J** et al. **An electrostatic repulsion model of centromere organization**. ***BioRxiv*** (2025)
12. Nakashima KN and Mihoubi FZ, […] **Huertas J**, Rubio-Sánchez R, Collepardo-Guevara R, O'Flaherty DK, Bonfio C. **Differential stability and dynamics of DNA-based and RNA-based coacervates affect non-enzymatic RNA chemistry** ***Nat. Comms*** (2025)
13. Chen L, [...] **Huertas J**, Redding S, Collepardo-Guevara R, Rosen MK. **Nucleosome Spacing Can Fine-Tune Higher Order Chromatin Assembly**. ***Nat. Comms*** (2025)
14. © Montez M and Zhu D, **Huertas J,** [...] Collepardo-Guevara R, Dean C. **Cold-induced nucleosome dynamics linked to silencing of Arabidopsis FLC.**. ***Nat. Comms*** **(**2025)
15. Maristany MJ, Aguirre A, Espinosa JR, **Huertas J**, Collepardo-Guevara R, Joseph JA. **Decoding Phase Separation of Prion-Like Domains through Data-Driven Scaling Laws**. ***Elife*** (2025)
16. Ng TLC and Hoare MP and Maristany MJ, Wilde EJ, Sneideris T, **Huertas J**, [...] Kumita JR. **Tandem-repeat proteins introduce tuneable properties to engineered biomolecular condensates.** ***ACS Chemical Science*** (2024)
17. Orsetti A and Van Oosten D, […] **Huertas J**, van Ingen H, Cojocaru V. **Structural dynamics in chromatin unraveling by pioneer transcription factors.** ***Biophysical Reviews*** (2024)
18. ★ **Huertas J**, Woods E, Collepardo-Guevara R, **Multiscale modelling of chromatin organisation: Resolving nucleosomes at near-atomistic resolution inside genes.** ***Current Opinion in Cell Biology***. (2022).
19. ★ **Huertas J**, Schöler HR, Cojocaru V. **Histone tails cooperate to control the breathing of genomic nucleosomes.** ***PLoS Computational Biology*** (2021).
20. ★ **Huertas J**, Cojocaru V. **Breaths, twists, and turns of atomistic nucleosomes.** ***J. Mol. Biol.*** (2021).
21. ★ **Huertas J,** MacCarthy CM, Schöler HR, Cojocaru V**. Nucleosomal DNA Dynamics Mediate Oct4 Pioneer Factor Binding.** ***Biophysical Journal*** (2020).
22. © Viplav A., Saha J, **Huertas J,** […] Galic M. **ArhGEF37 assists dynamin 2 during clathrin-mediated endocytosis**. ***J. Cell Sci.*** (2019).

GRANTS

FUNDING

2022 Herchel Smith Postdoctoral Research Fellowship (Awarded 2022, from 06/2023)

2022 HORIZON TMA Marie Skłodowska-Curie Action (MSCA) Postdoctoral Fellowship

COMPUTING GRANTS

2024 EPSRC High Performance Computing Grant. (114,000 CUs)

2018 Project extension (20 million hours). Gauss Call 2/2018

2017 Computing grant from the GAUSS CENTER (20 million hours). Project ID 12622. Gauss Call 2/2017

SUPERVISION AND TEACHING

SUPERVISION

Co-supervisor of a PhD Student in the Collepardo lab.

Co-supervisor of visiting students in the Collepardo lab.

(9 students, including undergraduates, Part II Genetics, MSc students, and summer interns)

TEACHING

Senior demonstrator Scientific Computing, Part IA (Department of Physics, University of Cambridge)

Lecturer Systems Biology, Part III (School of Biological Sciences, University of Cambridge)

Over 80 hours of personalized teaching (“Supervisions”).

(Mathematical Biology, Clare College and Emmanuel college, University of Cambridge)

Co-supervisor of 2 M.Sc students in the Schöler lab.

Practical Courses, M.Sc. Program (Biology Department, University of Münster)

Lectures and practical courses, M.Sc. program (Chemistry Department, University of Münster)

TALKS AND POSTERS

TALKS

25/02/2026 **2026 Annual Biophysical Society Meeting** (San Francisco, CA) ***Travel award**

15/02/2025 **2025 Annual Biophysical Society Meeting** (Los Angeles, CA)

27/11/2024 **Sociedad Argentina de Biofísica** (Bahía Blanca, Argentina)

24/06/2024 **International Union for Pure and Applied Biophysics 2024** (Kyoto, Japan)

19/06/2023 **Cambridge Epigenetics Club Summer Symposium** (Cambridge, UK)

09/05/2023 **4DGenomics2023: Investigating, modelling and understanding the genome in space and time** (Milano, Italy)

20/04/2023 **Multiscale simulations of DNA from electrons to nucleosomes: 22 years of the Ascona B-DNA Consortium** (Ascona, Switzerland)

20/02/2023 **2023 Biophysical Society Meeting** (San Diego, CA**)** ***Travel award / Poster prize**

16/11/2020 **Science for Life** (Utrecht – Virtual)

09/09/2020 **8th Annual German Stem Cell Network Conference** (Virtual)

21/07/2019 **Joint 12th EBSA 10th ICBP-IUPAP BIOPHYSICS CONGRESS** (Madrid)

26/04/2019 **Graduate School Meeting** (Münster, Germany)

16/11/2018 **Netherlands Society on Biomolecular Modelling meeting** (Utrecht)

21/06/2018 **ISQBP President meeting** (Barcelona)

POSTERS

23/02/2026 **2026 Annual Biophysical Society Meeting** (San Francisco, CA)

30/11/2025 **Sociedad Argentina de Biofísica** (Buenos Aires, Argentina)

19/02/2025 **2025 Annual Biophysical Society Meeting** (Los Angeles, CA)

29/11/2024 **Sociedad Argentina de Biofísica** (Bahía Blanca, Argentina)

12/02/2024 **2024 Annual Biophysical Society Meeting** (Philadelphia, PA) ***Travel award**

28/11/2023 **Sociedad Argentina de Biofísica** (Cordoba, Argentina)

04/04/2023 **5th Manchester Multiscale Conference** (Manchester, UK)

31/03/2023 **Physics of Emergent Behaviours IV Conference** (Harrogate, UK)

28/03/2023 **Physics of Life 2023** (Harrogate, UK)

01/04/2019 **Multiscale Modeling of Chromatin** (Les Houches, France) ***Poster Prize**

22/08/2018 **Biophysical Society Thematic Meeting** (Santa Cruz, CA)**,**

OTHER RELEVANT EXPERIENCE

Member of the Chromatin Consortium MBL

Visiting scientist in the **Marine Biology Lab in Woods Hole, Massachussets**.

An international collaborative initiative among leading research groups in the study of chromatin, including the groups of Mike Rosen (UT Southwestern), Vijay Ramani (UCSF), Bryan Gibson (St Jude's Hospital), Daniel Gerlich (IMBA,Vienna), Sy Redding (UMass) and Rosana Collepardo-Guevara (University of Cambridge)

Organizing member of the Early-Career Center for Physical Biology (Cambridge).

Organizer of the online Multiscale Genome Organization Talk Series.

Organizing member of the Young Biophysicists group – YIB under the Argentine Biophysics Society.

Organizer of the international CiM-IMPRS GRADUATE SCHOOL MEETING (2019).

Doctoral student representative at the Max Planck Institute for Molecular Biomedicine (2017–2021).

Organizer of various workshops for doctoral students, as part of the student representative initiative.

Experimental training in the laboratories of Prof. Justin Teissie (2011-2014) and Dr. Elena Gomez-Diaz (2013)